Evaluating the efficiency and water recovery of direct contact membrane distillation (DCMD) systems with dual heat recovery and recirculation of retentate and permeate

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Abstract

A transient model for hollow-fiber direct contact membrane distillation (DCMD) modules is employed to evaluate a desalination plant in six modes of operation, namely batch, semi-batch, volume-control, salinity-control, continuous, and steady state. The main goal is to reveal DCMD desalination configurations that present the best compromise between maximum energy efficiency and minimum use of brackish water from wells, mainly focusing on applications in sites under hydrological stress, like in the semi-arid Brazilian northeastern region. The obtained results indicate that using an additional heat exchanger for heat regeneration, combined with recirculation of the retentate stream, enhances the system efficiency compared to a scenario that uses only the distillate heat recovery. Additionally, the results also showed that if the main interest is in obtaining high recovery ratios (high volume of produced distillate water per volume of brackish water), the recirculating semi-batch and volume-controlling operating modes are the best alternatives to produce distil-

lated water. However, if the interest is to minimize the thermal energy cost, the best option is the batch operating mode.

Keywords: Desalination; DCMD system; heat recovery; retentate recirculation; energy efficiency.

Nomenclature

Latin Symbols (a–m)		Latin Symbols (n–z)	
Symbol	Description	Symbol	Description
a	Specific surface area	N	Molar flux
c_p	Specific heat at constant pressure	N_{fbr}	Number of fibers in the vessel
C	Thermal capacity	Nu	Nusselt number
C_s	Salt concentration	p	Pressure
d	Diameter	Pr	Prandtl number
D	Diffusivity / diffusion coefficient	q_c	Heat transferred to the cold sink
f	Friction factor	q_h	Heat transferred from the hot source
h	Convective heat transfer coefficient	q_{HX}	Heat transferred in the heat exchanger
h_{fg}	Enthalpy / heat of vaporization	Q	Rate of heat transfer
j_{wl}	Distillate flow rate	r	Radius
k	Thermal conductivity	R	Universal gas constant
L	Hollow-fiber module length	Re	Reynolds number
m	Mass	T	Temperature
$\dot{m}$	Mass flow rate	V	Volume
M	Molar mass	$\dot{V}$	Volumetric flow rate
		u	Stream velocity
		z	Position along the module
Greek Symbols		Subscripts / Superscripts	
Symbol	Description	Symbol	Description
α	Membrane effectiveness (salt)	a	Relative to air

Nomenclature (continued)

Acronyms			
Symbol	Description	Symbol	Description
AGMD	Air Gap Membrane Distillation	PGMD	Permeate Gap Membrane Distillation
CGMD	Conductive Gap Membrane Distillation	RO	Reverse Osmosis
DCMD	Direct Contact Membrane Distillation	RR	Recovery Ratio
DLR	Distillate Limiting Ratio	SEC	Specific Energy Consumption
GOR	Gained Output Ratio	SGMD	Sweeping Gas Membrane Distillation
FLR	Feed Limiting Ratio	V-AGMD	Vacuum Air Gap Membrane Distillation
HICO	Hot In, Cold Out	VMD	Vacuum Membrane Distillation
HOCI	Hot Out, Cold In		
HX	Heat Exchanger		
MD	Membrane Distillation		

1 Introduction

Freshwater scarcity, mainly due to population growth, has attracted the attention of researchers and industry for the development and improvement of techniques for the treatment, purification, and desalination of salt water. Among the main desalination techniques, those involving thermal separation methods, such as membrane distillation, require more energy due to the need to vaporize the water since the water vaporization enthalpy is about 2,400 kJ/kg [Sharqawy et al., 2010], while Gibbs free energy for seawater, which governs the energy use of the reverse osmosis method, is 2.7 kJ/kg [Elimelech and Phillip, 2011]. Therefore, one of the challenges that must be overcome for the application of thermal methods such as membrane distillation (MD) in water desalination, which aims to be competitive against the conventionally employed reverse osmosis (RO) process in small and medium-sized systems (< 100 m^3/day of water produced), is to minimize the demand for thermal energy, improving thermal efficiency.

The membrane distillation process enables the use of residual heat from other thermal processes or the heat available from renewable sources, such as solar collectors. However, heat losses by conduction through the membrane further increase the thermal consumption, which, when added to high-water consumption, limits the large-scale use of these systems [Khayet and Matsuura, 2011]. Nevertheless, according to Guan et al. [2015], when compared to conventional desalination processes, such as multi-stage flash distillation, multi-effect distillation, and reverse osmosis, the MD process has several benefits: low salinity sensitivity and high salt rejection (100% content, theoretically), low membrane vulnerability to fouling, greater performance under low severity operating conditions, and the viability of using low-grade waste heat from other thermal processes or renewable sources, such as solar collectors, as previously mentioned [Curcio and Drioli, 2005, Charcosset, 2009, Drioli et al., 2015].

Concerning its technology, according to the method of obtaining the vapor pressure gradient through the membrane, the MD is classified in the following configurations [Zhang et al., 2015, Swaminathan et al., 2018]: DCMD (Di-

rect Contact Membrane Distillation), AGMD (Air Gap Membrane Distillation), SGMD (Sweeping Gas Membrane Distillation), VMD (Vacuum Membrane Distillation), PGMD (Permeate Gap Membrane Distillation), CGMD (Conductive Gap Membrane Distillation), V-AGMD (Vacuum-Enhanced Air Gap Membrane Distillation), among other variants. Each configuration has its advantages and intrinsic disadvantages, so the choice of one of them is still a research theme, which is usually directed through metrics such as GOR (Gained Output Ratio) and SEC (Specific Energy Consumption), which are parameters for energy efficiency analysis. The DCMD is the most studied and straightforward configuration among them, widely used for desalination studies and concentration of aqueous solutions.

In general, the increase in the thermal efficiency of MD systems is achieved by improving the performance of membranes (by choosing membranes with low thermal conductivity, adequate porosity, pore size and pore size distribution, and thickness), as well as by reducing the amount of external heat to be supplied to the process [Gostoli and Sarti, 1998, Khayet et al., 2006, Al-Obaidani et al., 2008, Yao et al., 2013, Li et al., 2014, Swaminathan et al., 2018].

The heat supplied to the process can then be significantly reduced by the reuse of the heat available in the output currents of the desalination module through external heat exchangers, extending the applicability of the MD process and improving energy efficiency [Gilron et al., 2007, Lee et al., 2011, Hausmann et al., 2012, Lu and Chen, 2012, Zhang et al., 2015, Gonzalez-Bravo et al., 2015, Kim et al., 2015, Maheswari et al., 2015, Lin et al., 2014, **?**, Christie et al., 2020, Lisboa et al., 2021, Sampaio, 2022]. Obviously, operating conditions such as temperature, the feed and distillate flow rates, salinity of the feed stream, and its recirculation, among others, significantly affect the MD process and can improve or reduce process efficiency [Khayet et al., 2000, Wang et al., 2009, Zhang et al., 2010, Fan and Peng, 2012, Alkhudhiri et al., 2012, Guan et al., 2014].

Additionally, since MD presents a low recovery ratio per pass (ratio between the production rate of pure water and the feed flow), using some form of recircu-

lation of the brine/retentate stream is mandatory to increase efficiency. In this way, by considering a CGMD module, Swaminathan and Lienhard [2018] describe four operation strategies that enable high global recovery using a single-pass low recovery process (batch recirculation, semi-batch recirculation, continuous recirculation, and continuous multi-stage recirculation). The first two operations are classified as transient/discontinuous processes, while the last two are steady state ones. According to these authors, who studied the salinity variation from a lower limit of 70 g/kg to an upper limit of 260 g/kg, the batch operation is the best option since GOR is about twice for the same distillate flow when compared to the continuous recirculation operation.

Sampaio [2022] modeled a seawater desalination DCMD plant with heat recovery of the recirculating hot distillate stream by using a DCMD module composed by a bundle of hollow-fiber tubes. He combined analytical solutions for the mass transfer problem across the membranes with the finite volume method for solution the steady-state heat transfer problem. Still, based on Christie et al. [2020] and Lin et al. [2014], and to extend their works, Sampaio [2022] developed an approximation for the optimal value of the ratio between permeate and feed flow rates, and for the maximum mass recovery rate that, differently from the expressions developed by Christie et al. [2020] and Lin et al. [2014], take into consideration the membrane thermal efficiency.

Thus, based on the works of Swaminathan et al. [2018], Christie et al. [2020], Lisboa et al. [2021], and Sampaio [2022] and extending their analyses, the present work aims at the development of physical-mathematical modeling, in the unsteady-state regime, and subsequent numerical simulation of a hollow-fiber DCMD system with double recirculation and heat recovery. The main goal is firstly to evaluate the potential of heat recovery from the retentate stream to increase the energy efficiency of the system to produce distilled water. Furthermore, an application of the model to study the desalination process in regions with high seasonal hydrological stress, such as the semi-arid region of the Brazilian northeastern. In addition to the classical batch, semi-batch, continuous, and steady-state modes of operation, two intermediary modes, namely,

mixing tank volume controlling and mixing tank salt concentration controlling, are also introduced. The aim is to verify their effect on the behavior of the system efficiency and on the water recovery ratio (for minimizing the well brackish water consumption). These controlling schemes can also be operated in conjunction with total or partial recirculation, i.e., when the brine is rejected from the system with fraction rejections equal or different to the unity, respectively.

Table 1 presents, in a condensed way, a brief bibliographic review of the main experimental and simulation works on the thermal efficiency of MD systems, highlighting the main contributions of each of the works, especially, but not limited to, those works that adopt heat recovery and recirculation of flow streams using hollow fiber membrane configuration.

2 System Description

The MD desalination plant modelled in the present work is characterized by a water desalination system that operates in the transient regime, containing a DCMD module (a vessel with a bundle of hollow fiber membranes), two tanks (a mixing and collector), and running with dual recirculation and dual heat recovery, where the heat available at the output of the module from the feed and distillate streams are recovered through the heat exchangers, HX-1 and HX-2, respectively, as shown in Figure 1. The heat source and sink are constant-temperature heat reservoirs, so the feed and distillate streams reach the specified values of the source/sink temperatures by supplying/extracting exactly the corresponding amount of heat required.

At the initial time, the feed circuit is completely filled with brackish water at the well salinity (c_s^{well}) and ambient temperature (T_{amb}). The distillate circuit is filled with distilled water at the same ambient temperature. The feed stream leaves the mixing tank at state 3 (c_{s3}, $V_3 = V_{feed}$, T_3) and enters HX-1, which heats it to state 4 (c_{s4}, V_4, T_4) by receiving energy from the distillate leaving the DCMD module at state 11 (c_{s11}, V_{11}, T_{11}). The mandatory use of this heat

Table 1: Literature review on thermal efficiency in MD systems by heat recovery.

Reference	Heat Recovery			Observations
	Distillate	Distillate	Retentate	
Fane *et al.* (1987)	X	X	X	Experimental/Simulation – Hollow fiber – DCMD
Schneider *et al.* (1988)	X	X	X	Experimental – Hollow Fiber – DCMD
Kurokawa and Sawa (1996)	X	X	X	Simulation – Flat fiber – DCMD
Li and Sirkar (2004)	—	—	—	Experimental – Hollow Fiber – DCMD – Crossflow
Gilron *et al.* (2007)	X	X	X	Simulation – Hollow fiber – DCMD Cascade
Lee *et al.* (2011)	X	X	X	Experimental/Simulation – Hollow Fiber – DCMD Cascade
Hausmann *et al.* (2012)	—	X	X	Experimental/Simulation – Flat Fiber
He *et al.* (2013)	X	X	X	Simulation – Hollow fiber – DCMD Cascade
Chung *et al.* (2014)	X	X	X	Experimental/Simulation – Hollow fiber – DCMD cascade
Duong *et al.* (2015)	—	X	X	Experimental/Simulation – Flat Fiber – DCMD
Lin *et al.* (2014)	X	X	—	Simulation – DCMD – FLR/DLR
Guan *et al.* (2015)	X	X	X	Simulation – Hollow fiber – DCMD
Zhang *et al.* (2015)	X	X	X	Review on heat recovery in MD systems
Lokare *et al.* (2018)	—	X	X	Simulation – DCMD – Brine recycling
Christie *et al.* (2020)	X	X	X	Simulation – DCMD – FLR/DLR
Lisboa *et al.* (2021)	X	—	—	Simulation – Hollow fiber – DCMD
Sampaio (2022)	X	X	—	Simulation – Hollow fiber – DCMD

exchanger (HX-1) greatly increases the efficiency of the MD system by reducing the amount of heat (q_h) that must be supplied by the heater to raise the feed stream temperature to the working temperature at state 5 (c_{s5}, V_5, $T_5 = T_{feed}$).

The concentrated module effluent (the retentate) at state 6 (c_{s6}, V_6, T_6) can be removed from the system (as a reject stream) or it can be recycled

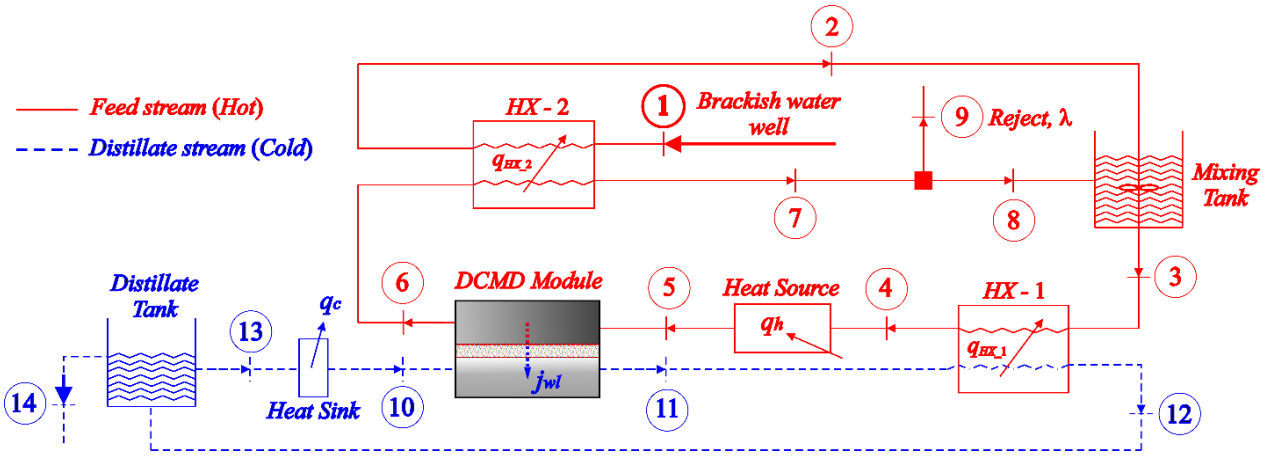


Figure 1: Schematic representation of the DCMD system with double heat recovery and partial or total recirculation of the retentate stream.

(recirculated). In the present work, the computational code was built taking into consideration both possibilities, simultaneously or not, depending on a flow rejection fraction (λ) at state 9 (c_{s9}, V_9, T_9), which directly and dynamically impacts the concentration of the solution in the tank and its volume, V_{mix}. When $\lambda = 1$, the brine is completely rejected at state 9 (the MD system must be continuously fed from the well on a permanent basis). Otherwise, when $\lambda < 1$, the retentate is partially recycled/rejected.

When necessary, depending on salinity and volume of the mixing tank (or the operating regime), brackish water can be pumped from the well at state 1 (V_{well}, c_s^{well}, T_{well}), passing through HX-2 as a cold stream to gain energy carried by the effluent retentate from the DCMD module at state 6, which will be cooled down to state 7 (c_{s7}, Q_7, T_7). The stream at state 7 can be directed, either totally ($\lambda = 0$) or partially ($0 < \lambda < 1$), to the mixing tank through state 8 (c_{s8}, V_8, T_8), to be mixed with the heated brackish water stream coming from state 2.

Thus, in addition to the desalination-system operating modes—batch, semi-batch, volume- or salinity-controlled, continuous, and steady-state operation—the numerical simulation will be used to investigate the energy efficiency of the system when an additional heat exchanger, HX-2, is employed. This heat exchanger enables thermal exchange between the retentate stream and the brack-

ish water supplied by the well. The simulation will also evaluate the partial rejection or recirculation of the retentate stream after heat recovery.

To extend the scope of the simulation, a parametric analysis will be performed to evaluate the interaction between the NaCl concentration in the feed stream and the amount of heat recovered from the fluid returning through the feed circuit of the desalination module. This analysis will also support the development of operational control strategies aimed at maximizing desalination efficiency while reducing brackish-water consumption from the well.To perform the present study, the operational ranges of the PLICA-7 DCMD module produced by Aquastill for experiments in a relevant environment are adopted.

3 Mathematical Model

The implemented mathematical model incorporates heat and mass transfer in all system components: mixing tank, distillate tank, DCMD module, heat source and sink, and heat exchangers 1 and 2. For the DCMD module, the model proposed by Lisboa et al. [2021] was extended to the unsteady-state regime. The authors treated the hollow-fiber DCMD module as an equivalent porous medium composed of the following constituents: the porous shell medium (where the feed stream flows), the porous lumen medium (where the distillate stream flows), and the membrane (where the distillation process occurs).An illustrative figure of the porous media model employed to analyze the heat and mass transfer in the vessel is shown in Figure 2.

The hollow-fiber membranes are assumed to be uniformly distributed over the vessel (module) cross-section, with their axes parallel to the vessel axis. The module porosity is defined as the ratio of the phase volume to the total vessel volume:

$$\varepsilon_\gamma = m + (-1)^m N_{\text{fbr}} \frac{r_\gamma^2}{r_v^2}, \tag{1}$$

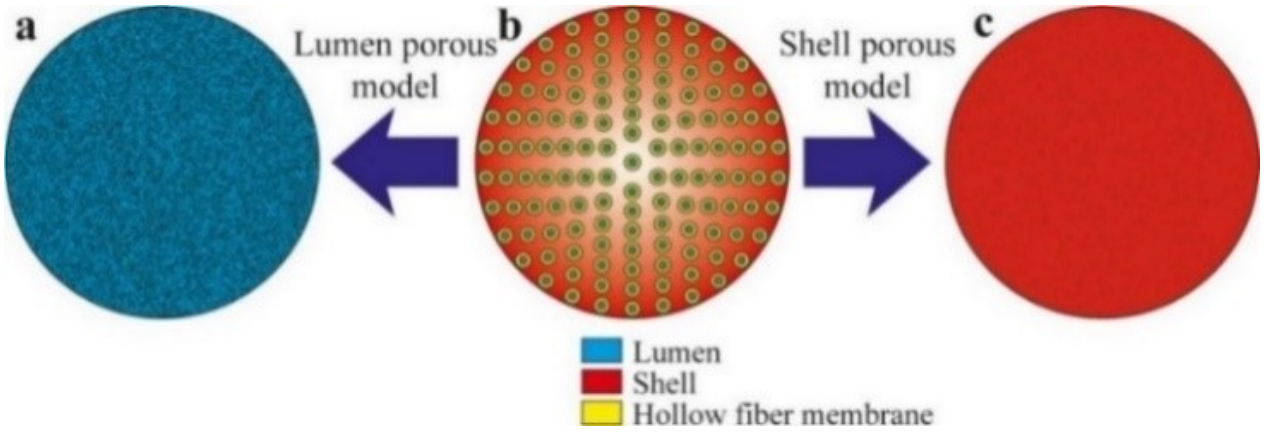


Figure 2: Illustration of the porous media models. Cross sections of the: (a) Porous medium, lumen side; (b) Module and hollow-fiber membrane bundle; (c) Porous medium, shell side. Figure 1 from Lisboa et al. [2021]. Reprinted with permission from Elsevier. Copyright 2021.

$$m = \begin{cases} 0, & \text{if } \gamma = l, \\ 1, & \text{if } \gamma = s. \end{cases} \tag{2}$$

$$\varepsilon_s = \frac{\pi r_v^2 - N_{fbr}\, \pi r_s^2}{\pi r_v^2} \tag{3}$$

Here, ε_γ is the porosity of phase γ (with $\gamma = l$ for the lumen phase and $\gamma = s$ for the shell phase), N_{fbr} is the number of fibers in the vessel, r_γ is the radius of the γ phase, and r_v is the radius of the vessel. Given that the membrane has a thickness δ, the outer radius of the membrane will be $r_s = r_l + \delta$.

The specific surface area, a_γ, used in heat and mass transfer calculations, is the ratio of the surface area available for transport phenomena in each phase to the vessel volume:

$$a_\gamma = \frac{N_{fbr}\, 2\pi r_\gamma}{\pi r_v^2} \tag{4}$$

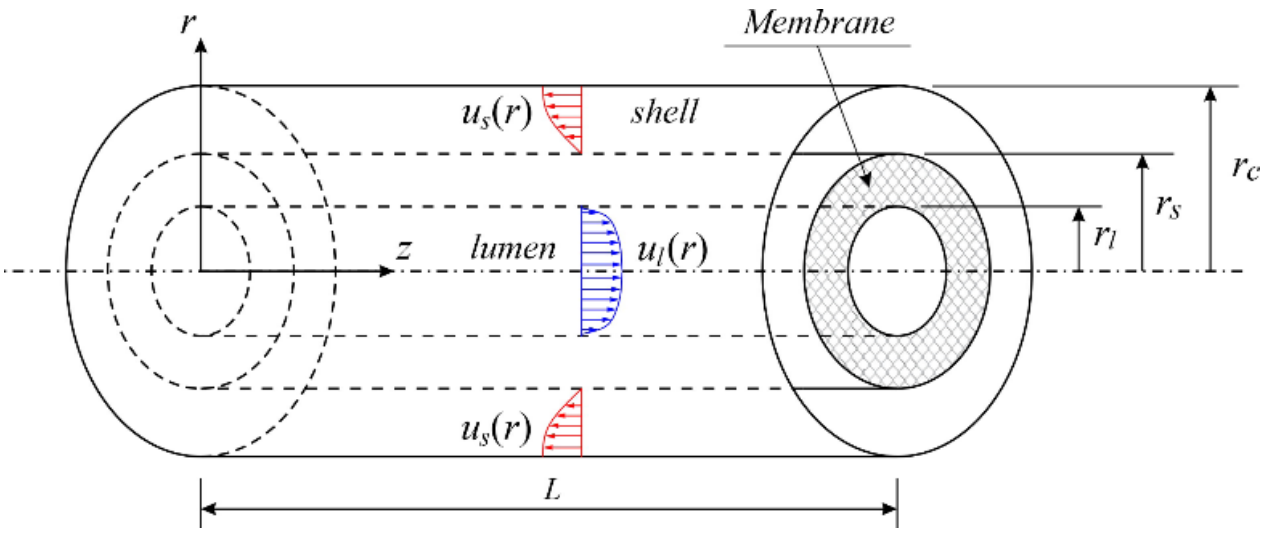


Figure 3: Representation for the single-fiber model of the hollow-fiber membrane module.

Considering that the heat exchangers, the distillate and mixing tanks, and the DCMD module are adiabatic, and that the salinity and temperature fields inside them are uniform (except inside the DCMD module), the physical situation is mathematically modelled as follows.

Mixture and Distillate Tanks

$$\frac{dV_{\text{mix}}}{dt} = \dot{V}_2 + \dot{V}_8 - \dot{V}_3, \qquad t > 0 \tag{5}$$

$$V_{\text{mix}}\frac{df_{\text{mix}}}{dt} = f_2\dot{V}_2 + f_8\dot{V}_8 - f_{\text{mix}}\left(\dot{V}_2 + \dot{V}_8\right), \qquad t > 0 \tag{6}$$

$$\rho_{\text{mix}}V_{\text{mix}}c_{p,\text{mix}}\frac{dT_{\text{mix}}}{dt} = \rho_2\dot{V}_2c_{p,2}T_2 + \rho_8\dot{V}_8c_{p,8}T_8 - c_{p,\text{mix}}T_{\text{mix}}\left(\rho_2\dot{V}_2 + \rho_8\dot{V}_8\right), \qquad t > 0 \tag{7}$$

$$\frac{dV_{\text{col}}}{dt} = \dot{V}_{12} - \dot{V}_{13} - \dot{V}_{14}, \qquad t > 0 \tag{8}$$

DCMD Module

$$\rho_l c_{p,l} \varepsilon_l \frac{\partial T_l}{\partial t} + \rho_l c_{p,l} u_l \frac{\partial T_l}{\partial z} = -a_l h_l \left(T_l - T_{m,l}\right), \qquad t > 0, \quad \text{lumen side} \tag{9}$$

$$\rho_s c_{p,s} \varepsilon_s \frac{\partial T_s}{\partial t} + \rho_s c_{p,s} u_s \frac{\partial T_s}{\partial z} = -a_s h_s \left(T_s - T_{m,s}\right), \qquad t > 0, \quad \text{shell side} \tag{10}$$

Equations (5)–(10) are subject to the following initial conditions:

$$t = 0: \qquad \begin{cases} V_{\text{mix}}(0) = V_{\text{mix},0}, \quad f_{\text{mix}}(0) = f_{\text{mix},0}, \quad T_{\text{mix}}(0) = T_{\text{amb}}, \\ V_{\text{col}}(0) = V_{\text{col},0}, \quad T_l(z,0) = T_{\text{amb}}, \quad T_s(z,0) = T_{\text{amb}}. \end{cases} \tag{11}$$

Additionally, since Equations (9) and (10) are partial differential equations, they are subject to the following boundary conditions:

$$T_l(0,t) = T_{e,l} = T_{l,0}, \qquad z = 0, \qquad t > 0 \tag{12}$$

$$T_s(L,t) = T_{e,s} = T_{s,0}, \qquad z = L, \qquad t > 0 \tag{13}$$

In these expressions, ρ_γ is the density of phase γ, $c_{p,\gamma}$ is the specific heat at constant pressure, k_γ is the thermal conductivity, T_γ is the bulk temperature, $T_{m,\gamma}$ is the interface temperature between the hollow fiber membrane surface and the liquid phase γ, and h_γ is the convection heat transfer coefficient. The convective heat transfer coefficients for the hot and cold streams are evaluated

through the following empirical correlations:

Heat-Transfer Correlations

The Nusselt number for phase γ is calculated as

$$Nu_\gamma = \frac{2r_\gamma h_\gamma}{k_\gamma} = \begin{cases} \left[4.36 + \dfrac{0.036\, Re_\gamma Pr_\gamma \left(d_{h,\gamma}/L\right)}{1 + 0.0011 \left[Re_\gamma Pr_\gamma \left(d_{h,\gamma}/L\right)\right]^{2/3}}\right] & \\ \quad \times \left[\dfrac{1 + N_{\text{fbr}} \left(r_s/r_v\right)}{1 - N_{\text{fbr}} \left(r_s/r_v\right)^2} \dfrac{r_s}{r_v}\right]^m, & Re_\gamma < 2100, \\ \dfrac{\left(Re_\gamma - 1000\right) Pr_\gamma \left(f_\gamma/8\right)}{1.07 + 12.7 \left(Pr_\gamma^{2/3} - 1\right) \left(f_\gamma/8\right)^{1/2}} & \\ \quad \times \left[\dfrac{1 + N_{\text{fbr}} \left(r_s/r_v\right)}{1 - N_{\text{fbr}} \left(r_s/r_v\right)^2} \dfrac{r_s}{r_v}\right]^m, & Re_\gamma \geq 2100. \end{cases} \tag{14}$$

Here, Re_γ is the Reynolds number based on the hydraulic diameter $d_{h,\gamma}$, Pr_γ is the Prandtl number, and f_γ is the friction factor. All these parameters are associated with phase γ and are defined as follows:

$$Re_\gamma = \frac{\rho_\gamma u_\gamma d_{h,\gamma}}{\mu_\gamma} \tag{15}$$

$$Pr_\gamma = \frac{\mu_\gamma c_{p,\gamma}}{k_\gamma} \tag{16}$$

$$d_{h,\gamma} = \begin{cases} 2r_l, & \gamma = l, \\ 2r_v \left[\dfrac{1 - N_{\text{fbr}} \left(r_s / r_v \right)^2}{1 + N_{\text{fbr}} \left(r_s / r_v \right)} \right], & \gamma = s. \end{cases} \tag{17}$$

$$f_\gamma = 2 \left[1.82 \log \left(Re_\gamma \right) - 1.64 \right]^{-2} \tag{18}$$

These equations are also coupled with the additional mass- and energy-balance equations for the remaining control volumes in the feed and distillate streams. Thus, the following relationships are obtained.

Temperatures

$$T_9 = T_8 = T_7 = \frac{1}{c_{p,7}} \left[c_{p,6} T_6 - \varepsilon_{\text{HX},2} \frac{C_{\min,2}}{\rho_6 \dot{V}_6} \left(T_6 - T_1 \right) \right] \tag{19}$$

$$T_{10} = T_{e,l}, \qquad \text{input} \tag{20}$$

$$T_{11} = T_l(L, t) \tag{21}$$

$$T_{14} = T_{13} = T_{12} = \frac{1}{c_{p,12}} \left[c_{p,11} T_{11} - \varepsilon_{\text{HX},1} \frac{C_{\min,1}}{\rho_{11} \dot{V}_{11}} \left(T_{11} - T_3 \right) \right] \tag{22}$$

Flow Rates

$$\dot{V}_1 = \dot{V}_2 = \dot{V}_{\text{well}}, \qquad \text{input} \tag{23}$$

$$\dot{V}_5 = \dot{V}_4 = \dot{V}_3 = \dot{V}_s, \qquad \text{input} \tag{24}$$

$$\dot{V}_7 = \dot{V}_6 = \left[\left(\varepsilon_m - 1\right) f_5 + 1\right] \dot{V}_5 - \dot{V}_{jwl} \tag{25}$$

$$\dot{V}_8 = \left(1 - l_{\text{rej}}\right) \dot{V}_7 \tag{26}$$

$$\dot{V}_9 = \dot{V}_{\text{rej}} = l_{\text{rej}} \dot{V}_7 \tag{27}$$

$$\dot{V}_{10} = \dot{V}_l, \qquad \text{input} \tag{28}$$

$$\dot{V}_{12} = \dot{V}_{11} = \dot{V}_{10} + \left(1 - \varepsilon_m\right) f_5 \dot{V}_5 + \dot{V}_{jwl} \tag{29}$$

$$\dot{V}_{13} = \left(1 - l_{\text{col}}\right) \dot{V}_{12} \tag{30}$$

$$\dot{V}_{14} = l_{\text{col}} \dot{V}_{12} \tag{31}$$

$$\dot{V}_{jwl} = \frac{j_{wl} A_{\text{memb}}}{\rho_{\text{H}_2\text{O}}} \tag{32}$$

Salinities

$$f_1 = f_2 = f_{\text{well}}, \qquad \text{input} \tag{33}$$

$$f_5 = f_4 = f_3 = f_{\text{mix}} \tag{34}$$

$$f_9 = f_8 = f_7 = f_6 = a_{\text{memb}} f_5 \frac{\dot{V}_5}{\dot{V}_6} \tag{35}$$

$$f_{10} = 0, \qquad \text{for } a_{\text{memb}} = 1, \qquad \text{input} \tag{36}$$

$$f_{14} = f_{13} = f_{12} = f_{11} = \frac{f_{10}\dot{V}_{10} + (1 - a_{\text{memb}})\, f_5 \dot{V}_5}{\dot{V}_{11}} \tag{37}$$

The temperatures T_4 and T_{12}, associated with HX-1, and the temperatures T_2 and T_7, associated with HX-2, are obtained from the definitions of the heat-exchanger effectivenesses:

$$\varepsilon_{\text{HX},1} = \frac{\dot{Q}_{\text{HX},1}}{\dot{Q}_{\text{HX},1,\text{max}}} = \frac{\rho_3 \dot{V}_3 \left(c_{p,4} T_4 - c_{p,3} T_3\right)}{C_{\text{min},1} \left(T_{11} - T_3\right)} = \frac{\rho_{11} \dot{V}_{11} \left(c_{p,11} T_{11} - c_{p,12} T_{12}\right)}{C_{\text{min},1} \left(T_{11} - T_3\right)}. \tag{38}$$

$$\varepsilon_{\text{HX},2} = \frac{\dot{Q}_{\text{HX},2}}{\dot{Q}_{\text{HX},2,\text{max}}} = \frac{\rho_1 \dot{V}_1 \left(c_{p,2} T_2 - c_{p,1} T_1\right)}{C_{\text{min},2} \left(T_6 - T_1\right)}$$

$$= \frac{\rho_6 \dot{V}_6 \left(c_{p,6} T_6 - c_{p,7} T_7\right)}{C_{\min,2} \left(T_6 - T_1\right)}. \quad (39)$$

The minimum heat-capacity rates of HX-1 and HX-2 are respectively given by

$$C_{\min,1} = \min \left\{ \rho_3 \dot{V}_3 c_{p,3},\ \rho_{11} \dot{V}_{11} c_{p,11} \right\} \quad (40)$$

$$C_{\min,2} = \min \left\{ \rho_1 \dot{V}_1 c_{p,1},\ \rho_6 \dot{V}_6 c_{p,6} \right\}. \quad (41)$$

In the salinity equations, α_{memb} represents the effectiveness of the membrane in retaining salt, that is, in preventing salt transport into the distillate stream. In the present study, complete salt retention is assumed:

$$\alpha_{\text{memb}} = 1. \quad (42)$$

The thermophysical properties of the brackish well water are evaluated at each position of the system as functions of the local temperature and salinity, according to the correlations proposed by Sparrow [2003].

The membrane-interface temperatures along the axial direction of the DCMD module, $T_{m,l}(z,t)$ and $T_{m,s}(z,t)$, required in Equations (9) and (10), are evaluated iteratively and dynamically within the numerical routine used to solve the system of differential equations. These temperatures are obtained by equating the local heat-transfer rates in the feed, membrane, and distillate regions.

The local convective heat-transfer rate from the shell-side stream to the membrane surface is

$$\dot{Q}_s = 2\pi r_s \Delta z \, h_s \left(T_s - T_{m,s}\right). \quad (43)$$

The total heat-transfer rate through the membrane is the sum of the conductive and latent heat-transfer rates:

$$\dot{Q}_{\text{memb}} = \dot{Q}_{\text{cond}} + \dot{Q}_{\text{vap}}. \tag{44}$$

The conductive heat-transfer rate through the cylindrical membrane is

$$\dot{Q}_{\text{cond}} = \frac{2\pi \overline{k}_m \Delta z}{\ln\left(r_s/r_l\right)} \left(T_{m,s} - T_{m,l}\right). \tag{45}$$

The latent heat-transfer rate associated with the water-vapor flux is

$$\dot{Q}_{\text{vap}} = 2\pi r_l \Delta z \, j_{wl} h_{fg}. \tag{46}$$

The local convective heat-transfer rate from the membrane surface to the lumen-side stream is

$$\dot{Q}_l = 2\pi r_l \Delta z \, h_l \left(T_{m,l} - T_l\right). \tag{47}$$

By combining Equations (43)–(47), the shell-side membrane interface temperature is obtained as

$$T_{m,s} = T_s - \left[1 + \frac{r_s h_s \left(T_{m,s} - T_{m,l}\right) \ln\left(r_s/r_l\right)}{\overline{k}_m \left(T_{m,s} - T_{m,l}\right) - j_{wl} h_{fg} r_l \ln\left(r_s/r_l\right)}\right]^{-1} \times \left(T_s - T_l\right). \tag{48}$$

Similarly, the lumen-side membrane interface temperature is given by

$$T_{m,l} = T_l + \left[1 + \frac{r_l h_l \left(T_{m,s} - T_{m,l}\right) \ln\left(r_s/r_l\right)}{\overline{k}_m \left(T_{m,s} - T_{m,l}\right) - j_{wl} h_{fg} r_l \ln\left(r_s/r_l\right)}\right]^{-1} \times \left(T_s - T_l\right). \quad (49)$$

The density of the mixture at each position or control volume of the circuit, ρ_i, is calculated from the local salt mass fraction, f_i, using the densities of pure water and salt at $T_{\text{amb}} = 25\,^{\circ}\text{C}$. The reference values employed are $\rho_{\text{salt}} = 2165\ \text{kg}\,\text{m}^{-3}$ and $\rho_{\text{H}_2\text{O}} = 997.4\ \text{kg}\,\text{m}^{-3}$. Thus,

$$\rho_i = \left(1 - f_i\right) \rho_{\text{H}_2\text{O}} + f_i \rho_{\text{salt}}. \quad (50)$$

Other thermophysical properties, including the specific heat at constant pressure, thermal conductivity, dynamic viscosity, and enthalpy of vaporization, are also evaluated as functions of salinity and temperature. These properties are calculated using the correlations proposed by Sparrow [2003] or by interpolation between values available in thermophysical-property tables. The model used to calculate the effective thermal conductivity of the membrane, $\overline{k}_m$, considering both the polymer matrix and the water vapor contained within the membrane pores, is presented subsequently.

3.1 Modeling Mass Transfer in the Membrane

To partially close the system of equations, the mass-transfer process inside the DCMD module must be modeled to determine the water-vapor mass flux through the membrane, j_{wl}. Following the procedure adopted by Lisboa et al. [2021], the Dusty Gas Model proposed by **?**, one of the most widely used models for predicting distilled-water transport through porous membranes, is employed.

The molar transport of water vapor through the membrane pores is described by

$$\frac{N_w}{D_w^k} + \frac{p_a N_w - p_w N_a}{D_{wa}^0} = -\frac{1}{RT_m}\nabla p_w, \tag{51}$$

where N_w is the molar flux of water vapor, N_a is the molar flux of air, p_a is the partial pressure of air inside the membrane pores, p_w is the partial pressure of water vapor, D_w^k is the Knudsen diffusivity of water vapor, D_{wa}^0 is the molecular diffusivity of water vapor in air, R is the universal gas constant, and T_m is the average thermodynamic temperature at each axial position along the membrane.

Air is assumed to remain stationary inside the membrane pores, such that

$$N_a \approx 0. \tag{52}$$

Applying Dalton's law, the molar flux of water vapor through the hollow-fiber membrane, based on the inner membrane radius, is given by

$$N_{wl} = \frac{D_{wa}^0}{RT_m r_l \ln\left(r_s/r_l\right)} \ln\left[\frac{D_{wa}^0 - D_{\text{eff}} p_s}{D_{wa}^0 - D_{\text{eff}} p_l}\right]. \tag{53}$$

The corresponding water-vapor mass flux is

$$j_{wl} = M_w N_{wl}, \tag{54}$$

where M_w is the molar mass of water.

The molecular diffusivity of water vapor in air is calculated from

$$D_{wa}^0 = 4.46 \times 10^{-6} \frac{\varepsilon_m}{\tau_m} T_m^{2.334}. \tag{55}$$

The Knudsen diffusivity of water vapor is given by

$$D_w^k = \frac{\varepsilon_m}{\tau_m}\frac{d_p}{3}\sqrt{\frac{8RT_m}{\pi M_w}}. \tag{56}$$

The effective diffusivity resulting from the combined molecular and Knudsen transport mechanisms is

$$D_{\text{eff}} = \frac{D_w^k D_{wa}^0}{D_{wa}^0 + P D_w^k}. \tag{57}$$

The total pressure inside the membrane pores, P, is assumed to be equal to atmospheric pressure. The water-vapor partial pressures at the shell-side and lumen-side membrane interfaces, p_s and p_l, respectively, are calculated from the corresponding saturation vapor pressures. The effect of salinity is included through the salt mole fraction, $x_{\text{salt},\gamma}$, and the water activity coefficient, a_γ, according to

$$p_\gamma = (1 - x_{\text{salt},\gamma})\, a_\gamma p_{v,\gamma}, \qquad \gamma \in \{s, l\}, \tag{58}$$

where $p_{v,\gamma}$ is the saturation vapor pressure evaluated at the corresponding membrane-interface temperature using the Antoine equation.

The activity coefficient is calculated as

$$a_\gamma = 1 - 0.5 x_{\text{salt},\gamma} - 10 x_{\text{salt},\gamma}^2. \tag{59}$$

The morphological characteristics of the membrane are included in the calculation of the distillate mass flux through the mean pore diameter, d_p, porosity, ε_m, and tortuosity, τ_m.

Models for the effective membrane thermal conductivity, $\overline{k}_m$, and the membrane tortuosity, τ_m, must also be selected to close the governing equations.

The most common effective-thermal-conductivity models are based on different geometrical arrangements of the solid and gaseous phases. In the parallel model, the solid and gaseous phases are arranged in parallel:

$$\overline{k}_m = (1-\varepsilon_m)\,\overline{k}_s + \varepsilon_m \overline{k}_f, \qquad \text{parallel model.} \tag{60}$$

In the series model, the two phases are arranged in series:

$$\overline{k}_m = \left[\frac{1-\varepsilon_m}{\overline{k}_s} + \frac{\varepsilon_m}{\overline{k}_f}\right]^{-1}, \qquad \text{series model.} \tag{61}$$

For porous media with porosity greater than approximately 0.6, **?** recommend the Maxwell I model. This model assumes that the polymeric matrix consists of a random distribution of solid particles that do not interact with one another within a homogeneous fluid medium. The resulting effective conductivity is

$$\overline{k}_m = \overline{k}_f \left[\frac{1+2\,(1-\varepsilon_m)\,b_{sf}}{1-(1-\varepsilon_m)\,b_{sf}}\right], \qquad \text{Maxwell I model,} \tag{62}$$

where

$$b_{sf} = \frac{\overline{k}_s - \overline{k}_f}{\overline{k}_s + 2\overline{k}_f}. \tag{63}$$

The manner in which the pores and solid matrix are distributed within the membrane governs the diffusion path and, consequently, the vapor flux. Membrane tortuosity, defined as the ratio between the actual diffusion path length and the membrane thickness, is therefore an important parameter for predicting the transmembrane mass flux.

Several tortuosity models are available in the literature [Ghanbarian et al.,

2013]. Kim et al. [2021] analyzed eleven tortuosity models, including eight geometrical models based on Euclidean structures, such as cubes, spheres, and cylinders, and three models based on the fractal theory of porous media.

For the geometrical model proposed by **?**, the membrane tortuosity is

$$\tau_m = \frac{(2-\varepsilon_m)^2}{\varepsilon_m}. \tag{64}$$

The model proposed by Beeckman [1990] is

$$\tau_m = \frac{\varepsilon_m}{1-(1-\varepsilon_m)^{1/3}}. \tag{65}$$

For the fractal model proposed by Elias-Kohav et al. [1991], the tortuosity is

$$\tau_m = \frac{1}{\varepsilon_m}. \tag{66}$$

The Bruggeman model is given by

$$\tau_m = \frac{1}{\sqrt{\varepsilon_m}}. \tag{67}$$

According to **?**, who evaluated several effective-thermal- conductivity and tortuosity models for membrane-distillation simulations in flat-sheet and hollow-fiber configurations, an appropriate model should be selected based on the known morphological and structural properties of the membrane employed in the experimental desalination module.

3.2 Efficiency Metrics

To evaluate the desalination system and quantify improvements in the energy efficiency of the membrane-distillation process, the thermal efficiency, η_{th}, gain-output ratio, GOR, specific thermal energy consumption, SEC_{th}, and water recovery ratio, RR, are calculated under the different operating scenarios.

The thermal efficiency of the membrane is defined as

$$\eta_{th} = \frac{\dot{Q}_{\mathrm{vap}}}{\dot{Q}_{\mathrm{memb}}}. \tag{68}$$

The gain-output ratio is defined as

$$\mathrm{GOR} = \frac{\dot{Q}_{\mathrm{vap}}}{\dot{Q}_{\mathrm{ext}}}. \tag{69}$$

The specific thermal energy consumption is

$$\mathrm{SEC}_{th} = \frac{\dot{Q}_{\mathrm{ext}}}{\dot{V}_{jwl}}. \tag{70}$$

The water recovery ratio is

$$\mathrm{RR} = \frac{\dot{m}_{\mathrm{dist}}}{\dot{m}_{\mathrm{well}}}. \tag{71}$$

The total heat-transfer rate through the membrane is the sum of the conductive and latent heat-transfer rates:

$$\dot{Q}_{\mathrm{memb}} = \dot{Q}_{\mathrm{cond}} + \dot{Q}_{\mathrm{vap}}. \tag{72}$$

The total conductive heat-transfer rate through all hollow fibers is calculated

as

$$\dot{Q}_{\text{cond}} = \left| N_{\text{fbr}} \int_0^L \frac{2\pi \overline{k}_m}{\ln\left(r_s/r_l\right)} \left(T_{m,s} - T_{m,l}\right) \, \mathrm{d}z \right| . \tag{73}$$

The total latent heat-transfer rate associated with water-vapor transport is

$$\dot{Q}_{\text{vap}} = \left| N_{\text{fbr}} \int_0^L 2\pi r_l j_{wl} h_{fg} \, \mathrm{d}z \right| . \tag{74}$$

The external heat demand is calculated by subtracting the heat recovered in HX-1 and HX-2 from the total thermal demand:

$$\dot{Q}_{\text{ext}} = \dot{Q}_{\text{tot}} - \left(\dot{Q}_{\text{HX},1} + \dot{Q}_{\text{HX},2} \right) . \tag{75}$$

The total thermal demand required to heat the feed stream is

$$\dot{Q}_{\text{tot}} = \rho_3 \dot{V}_3 c_{p,3} \left(T_{e,s} - T_{e,l}\right) . \tag{76}$$

The heat-transfer rate recovered in HX-1 is

$$\begin{aligned} \dot{Q}_{\text{HX},1} &= \rho_3 \dot{V}_3 \left(c_{p,4} T_4 - c_{p,3} T_3\right) \\ &= \rho_{11} \dot{V}_{11} \left(c_{p,11} T_{11} - c_{p,12} T_{12}\right) . \end{aligned} \tag{77}$$

The heat-transfer rate recovered in HX-2 is

$$\begin{aligned} \dot{Q}_{\text{HX},2} &= \rho_1 \dot{V}_1 \left(c_{p,2} T_2 - c_{p,1} T_1\right) \\ &= \rho_6 \dot{V}_6 \left(c_{p,6} T_6 - c_{p,7} T_7\right) . \end{aligned} \tag{78}$$

The global GOR and SEC_{th} values are calculated as average values over the complete simulation period. The volumes of distilled water produced and brackish well water consumed are obtained by integrating the corresponding instantaneous volumetric flow rates over time.

A computational code for solving the coupled system of ordinary and partial differential equations governing the temperatures, flow rates, and salinities was developed in Fortran 95. The MMOCLH routine [IMSL Fortran Math Library, 2022], which solves hyperbolic partial differential equations using the method of lines, was employed with an error tolerance of 10^{-5}.

The nonlinear equations used to calculate the membrane-interface temperatures, $T_{m,s}$ and $T_{m,l}$, were solved with a prescribed error tolerance of 10^{-11} using the ZBREN routine [IMSL Fortran Math Library, 2022].

4 The System Operating Modes

To control the operation of the dynamic system and optimize the desalination system water production and brackish water consumption, a general algorithm based on volume and salinity of the mixture in the mixing reservoir is applied at each instant of time. The developed computational code is very flexible, allowing simulations of different desalination system operation scenarios: steady-state regime (no recirculation), batch recirculation, semi-batch recirculation, continuous recirculation (dynamic rejection fraction/recirculation), and partial recirculation operation mode with volume or concentration control.

For the unsteady-state regimes, the system can start the operation with no brine rejection (total recirculation), $\lambda = \lambda_0 = 0$, or with a predefined rejection (partial recirculation), $\lambda_0 \neq 0$. For these regimes, the desalination system operates in such a way that the mixing tank, initially with a volume of brackish water of $V_{mix,0} = 100$ L, has its mixture consumed by the DCMD module, as distillate is produced over time.

Additionally, to analyze the influence of the external heat exchangers on the

system performance it is sufficient to vary their effectiveness (ε_{HX1} and ε_{HX2}). By choosing $\varepsilon_{HX1} = 0$ and/or $\varepsilon_{HX2} = 0$, the heat exchangers are not included in the analysis.

In batch and semi-batch operation modes, over each process cycle, the distillate is continuously removed from the system, and the remaining solution that leaves the MD module, with higher salinity, is directed to the feed tank ($\lambda = 0$). In the batch operation process, there is no water pumping from the well ($\dot{V}_{well} = 0$) between recycling, so the volume of the solution in the tank decreases, and the salt concentration increases over time. In the semi-batch operation cycle, the volume of the mixing tank is kept constant, by continuously adding fresh feed water from the well to the system, at the same rate at which distillate is removed ($\dot{V}_{well} = \dot{V}_{jwl}$).

In both operation processes, the permeate production rate and the heat-transfer rate decrease as the feed stream of the MD module becomes increasingly saturated with salts. After the salinity of the solution in the mixing tank reaches a limiting value, the high-salinity brine is flushed-out (λ is set to 1) until the minimum allowable volume is reached. Then, the well pump refills the tank from the well ($\dot{V}_{well} = \dot{V}_s$) until the minimum concentration is reached, and the cycle restarts.

In the continuous operation regime, a fixed operational brine salinity is specified at the MD module exit. Therefore, after the initial transients have elapsed and the solution that leaves the module is at the required salinity (reaching the steady-state regime), it must be continuously withdrawn from the system at a specific fraction rejection (λ) that depends on the MD module performance. The make-up water from the well ($\dot{V}_{well}$) is simultaneously added to the tank at a proportional flow rate that maintains the salinity of the stream at the MD module exit, which is associated with the corresponding rejection fraction.

The rejection fraction and the volumetric flow rate from the well are obtained from the following salt and total-mass balances applied to the mixing tank:

$$\lambda = 1 - \frac{\dot{V}_{jwl}\, C_{s,well}}{\left(\dot{V}_s - \dot{V}_{jwl}\right)\left(C_{s6} - C_{s,well}\right)} \tag{79}$$

$$\dot{V}_{well} = \dot{V}_{jwl} + \lambda\left(\dot{V}_s - \dot{V}_{jwl}\right) \tag{80}$$

To simulate a steady-state condition with no recirculation, it is sufficient to set $\lambda = \lambda_0 = \lambda_1 = 1$ and $\dot{V}_{well} = \dot{V}_s$ all the time during the simulations. In this situation, all retentate leaving the DCMD module is rejected and the system is continuously fed by the well at constant feed flow rate and salinity.

In the mixing-tank volume-control operating mode, minimum and maximum allowable tank volumes are specified as operating limits. The salinity of the water in the mixing tank ($C_{s,mix}(t)$) is continuously increasing with the distillate production, and its volume ($V_{mix}(t)$) decreases. When the minimum volume is reached (V_{min}), the tank is filled with brackish water from the well up to the maximum volume (V_{max}) previously chosen. The process is then repeated until the desalination operation is completed.

In the mixing-tank salinity-control operating mode, minimum and maximum allowable salinities are specified as operating limits. The salinity of the water in the mixing tank ($C_{s,mix}(t)$) is continuously increasing with the distillate production, while its volume ($V_{mix}(t)$) decreases since there is no brackish water flow from the well. When the maximum allowable salinity is reached ($C_{s,max}$), the pump from the well is turned on to feed the mixing tank until the salinity of the water contained in it falls to the minimum predefined value.

5 Results

Before analyzing typical well operation scenarios, the developed computational code is verified by comparing its results with available literature data. Since

no transient results are available, these comparisons are made with data for the steady-state regime.

Table 2 presents the physical and geometric characteristics of the membranes and modules employed in the comparisons. The experimental results of Wang et al. [2008], Yang et al. [2011] and numerical results of Guan et al. [2014] and Kim [2014] are employed for situations with no heat exchangers.

Table 2: Characteristics (input data) of the membranes and modules used for comparison.

Parameter	Symbol	Unit	Wang *et al.* (2008)	Kim (2014)	Yang *et al.* (2011)	Guan *et al.* (2014)
			Experimental	Numerical	Experimental	Numerical
Membrane inner radius	r_l	mm	0.3	0.3	0.46	0.49
Membrane thickness	t	mm	0.11	0.11	0.275	0.240
Polymer thermal conductivity	k_p	W/(m·K)	0.18[a]	0.18	0.18[a]	0.18[a]
Average pore diameter	d_p	mm	0.17[b]	0.17	0.145/0.191	0.2[b]
Porosity	ε_m	—	0.738	0.738	0.83	0.85[b]
Shell inlet temperature	$T_{e,s}$	°C	30–79.3	30–79.3	35–70	30–80
Lumen inlet temperature	$T_{e,l}$	°C	17.5	17.5	25	23.5–30
Total pore pressure	P	kPa	101.325[a]	101.325[a]	101.325[a]	101.325[a]
Shell side flow rate	$\dot{V}_s$	L/min	1.648[c]	1.521	2.5–3.0	1.14–3.0
Lumen side flow rate	$\dot{V}_l$	L/min	0.407[c]	0.407	0.4	0.046–0.4
Number of fibers	N_{fbr}	—	30[a]	30	20	22
Vessel radius	r_v	mm	3.241[a]	3.232	4.75	7.5
Vessel length	L	m	0.200	0.200	0.225	0.200
Salt mass fraction shell	$w_{NaCl,s}$	%	3.5[d]	3.5	3.5	7.0
Lumen salt mass fraction	$w_{NaCl,l}$	%	0.0	0.0	0.0	0.0
Cold/hot fluid arrangement	—	—	HOCI	HOCI	HOCI	HICO
Direction of flow	—	—	Co-current	Counter	Counter	Counter
Membrane area	A_{memb}	cm^2	150	152.7	220	180.9
Packing density	φ	%	48[a]	48	50	20.8[a]

[a] Obtained from, or employed the calculation procedure in, Kim [2014]. [b] Assumed in comparison with other works. [c] Calculated as a function of average velocities. [d] Lisboa et al. [2021] used $w_{NaCl,s} = 2.5\%$ in their simulations.

Table 3 presents a verification of the results obtained with the present work together with the work of Lisboa et al. [2021] and a comparison with the experimental results of Wang et al. [2008].

The results show that the present methodology is consistent with the an-

Table 3: Verification of the computational code with the numerical results of Lisboa et al. [2021] and comparison with the experimental data of Wang et al. [2008]. $T_{e,l} = 17.5$°C.

$T_{e,s}$ [°C]	ΔT [°C]	$T_{o,s}$ [°C]		$T_{o,l}$ [°C]		j_{wl} [kg/(m^2·h)]	
		Present	Wang (2008)	Present	Wang (2008)	Present	Wang (2008)
40.0	22.5	38.5	39.0	23.4	24.5	5.5	6.2 ± 0.4
49.5	23.0	47.2	48.0	26.5	27.8	10.1	11.5 ± 0.7
59.8	42.3	56.2	57.4	31.7	31.3	20.0	19.9 ± 1.0
70.2	52.7	65.1	67.2	37.5	36.2	32.8	30.8 ± 1.0
79.3	61.8	72.6	76.0	43.8	39.0	47.9	41.5 ± 1.2

* Based on the outer radius of the hollow fiber. $T_{Wang_et_al.}$(2008): $T \pm 0.1$°C.

alyzed physics, at least in the steady-state regime, when compared to the experimental results of Wang et al. [2008]. This table also shows that the results obtained with the methodology in transient regime converge, as would be expected for long times, to the numerical results of Lisboa et al. [2021].

Another verification is illustrated in Figure 4, which shows a comparison of the results obtained in the present work for the produced distillate flux with the experimental results of Wang et al. [2008] and the numerical results of Kim [2014].

To illustrate the spatial temperature distribution along the DCMD module, Figure 5 brings a comparison between the results obtained with the present formulation and the numerical results of Guan et al. [2014] for the steady-state temperatures along the feed channel, T_s, of the feed/membrane interface, $T_{m,s}$, of the membrane/distillate interface, $T_{m,l}$, and of the distillate channel, T_l. The inlet temperatures and flow rates of the feed and distillate streams employed in the simulation were: $T_{e,s} = 80$°C, $T_{e,l} = 30$°C and $\dot{V}_s = 1.1425$ L/min, $\dot{V}_l = 0.04635$ L/min, respectively.

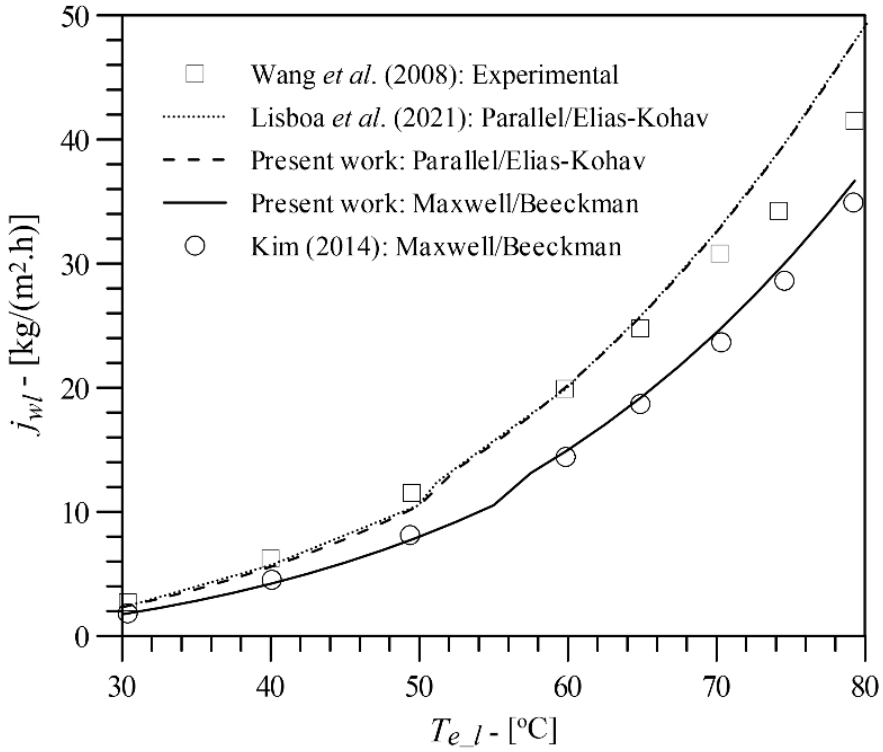


Figure 4: Comparison between the result obtained with the present methodology and the experimental results of Wang et al. [2008] and numerical results of Kim [2014] for the distillate flux.

5.1 Simulations with the Brackish Water Well

For simulation of a real desalination system fed by brackish water wells, data from wells commonly found in the northeastern region of Brazil were used, whose temperatures, salt concentrations, typical flow rates, and other properties can be found in Veiga [2017]. Table 4 provides the characteristic parameter values of 444 wells located in the Ceará state of Brazil.

The parameters relating to the characteristics of the DCMD module and the membrane are those provided in Table 5. The flow arrangement of the module is carried out in the HOCI configuration (hot fluid in the shell and cold fluid in the lumen) in counter-current.

To illustrate a typical dynamic behavior of the heat transfer process inside the DCMD module, Figure 6 shows the bulk temperature profiles of the retentate and distillate streams along the respective channels inside the module, at

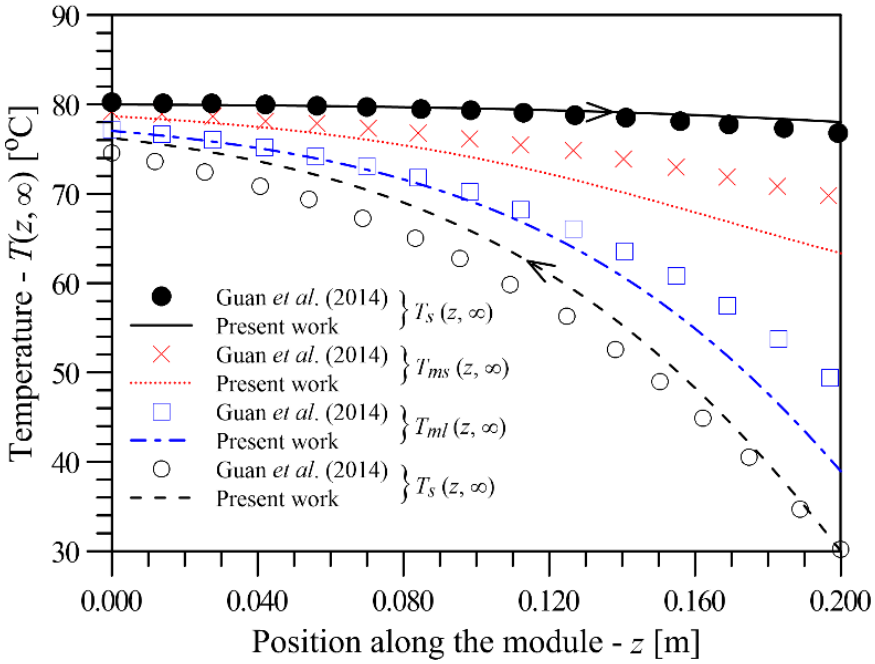


Figure 5: Temperature distribution along the axial position in the DCMD module in the feed, interface, and distillate regions. Comparison with the numerical results of Guan et al. [2014].

Table 4: Characteristics of wells in the state of Ceará/Brazil.

Limits	Salt concentration [g_s/L]	Flow rates [m^3/h]
Maximum	13	15
Minimum	0.199	0.10
Average	2.532	3.45
Adopted	3.5	3

$T_{well} = T_{amb} = 25°C$

four instants of time (approximately 2.8×10^{-4} s, 3.4 s, 16 s, and 30 s), for the flow condition $\dot{V}_l = \dot{V}_s = 50$ L/min.

As can be seen from this figure, the time evolution of the heat transfer inside the DCMD module is very fast, since the steady-state condition is reached in just a few seconds, taking no more than 30 s for the module to attain this regime.

Figures 7a and 7b illustrate bulk temperature profiles of the feed and per-

Table 5: Input data for the membrane and module used in the well simulations.

Parameter	Symbol	Unit	Lisboa *et al.* (2021)
Membrane inner radius	r_l	mm	0.3
Membrane thickness	t	mm	0.1
Polymer thermal conductivity	k_p	W/(m·K)	0.2
Average pore diameter	d_p	mm	0.2
Membrane porosity	ε_m	—	0.8
Shell-side inlet temperature	$T_{e,s}$	°C	80
Lumen-side inlet temperature	$T_{e,l}$	°C	25
Total pore pressure	P	kPa	101.325
Shell-side flow rate	$\dot{V}_s$	L/min	30–50–80
Lumen-side flow rate	$\dot{V}_l$	L/min	30–50–80
Number of fibers	N_{fbr}	—	10,000
Vessel radius	r_v	mm	50
Vessel length	L	m	1
Shell-side salt mass fraction	$w_{NaCl,s}$	%	0.035
Lumen-side salt mass fraction	$w_{NaCl,l}$	%	0
Membrane area	A_{memb}	m^2	7.2
Packing density	φ	%	0.64

meate channels along the module for different conditions of volumetric flow rate of the streams, considering steady-state single-pass flow in the DCMD module, and no heat recovery ($\lambda_0 = \lambda = 1$, $\dot{V}_{well} = \dot{V}_s$, $\varepsilon_{HX1} = \varepsilon_{HX2} = 0$).

For an intuitive understanding of how the operational and configurational decisions affect energy efficiency, Christie et al. [2020], based on Lin et al. [2014], introduced the critical mass-flow-rate ratio, α^*, to define the theoretical condition for the maximum water-recovery factor in steady-state, single-pass DCMD systems, $R_{\max}$:

$$\alpha^* = \left.\frac{\dot{m}_s\, c_{p,s}}{\dot{m}_l\, c_{p,l}}\right|_{\text{critical}} \tag{81}$$

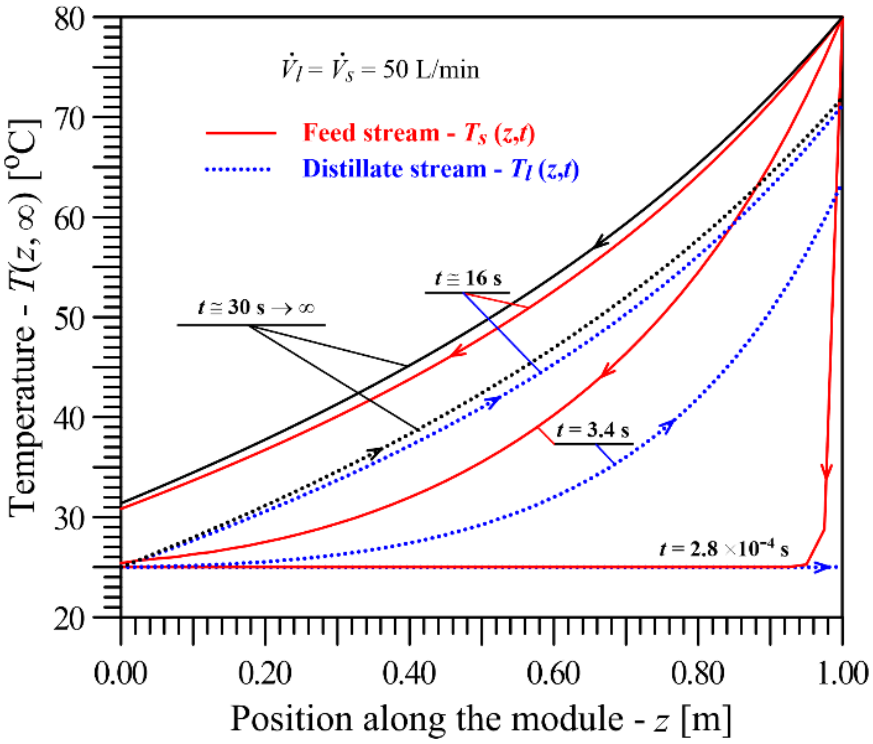


Figure 6: Time evolution of temperature profiles inside the DCMD module for $\dot{V}_l = \dot{V}_s = 50$ L/min.

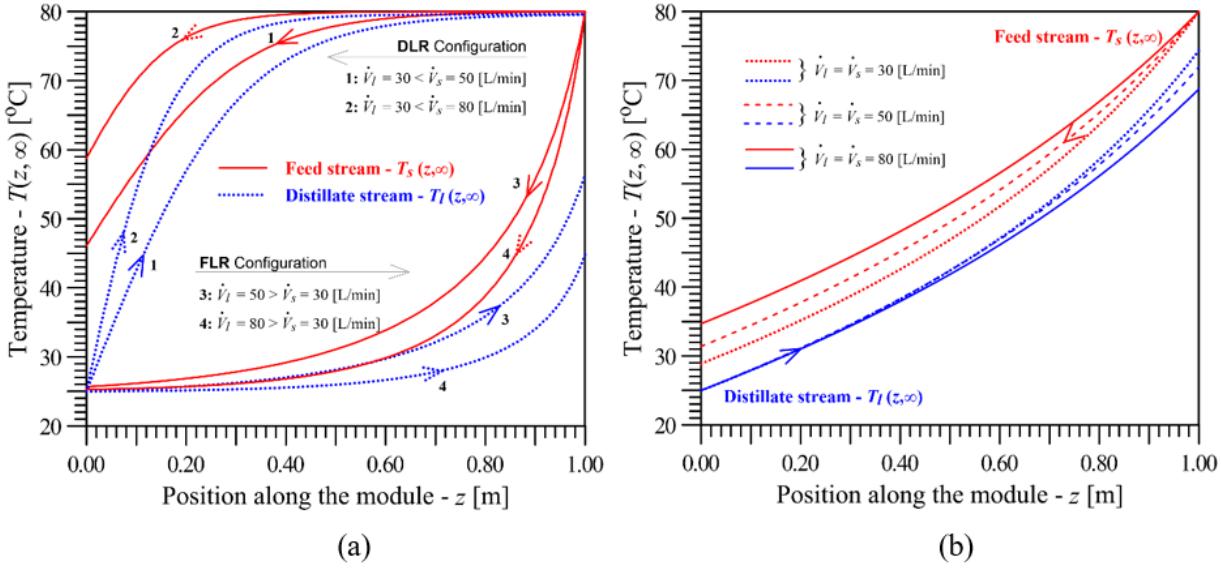


Figure 7: Steady-state temperature profiles along the axial position in the module DCMD for: a) different feed and distillate flow rates; b) equal distillate and feed flow rates ($\dot{V}_l = \dot{V}_s$).

$$\alpha = \frac{\dot{m}_s \, c_{p,s}}{\dot{m}_l \, c_{p,l}} \tag{82}$$

$$R_{\max}^{FLR} > R_{\max}^{DLR} \tag{83}$$

For $\alpha < \alpha^*$, the module operates in the DLR regime. Conversely, for $\alpha > \alpha^*$, the module operates in the FLR regime.

5.1.1 Operating Regime Analysis

After characterizing the DCMD desalination system, its behavior under different operating modes is investigated. In all cases, the FLR configuration is adopted with $\dot{V}_l = \dot{V}_s = 50 \text{ L min}^{-1}$.

Table 6 summarizes the averaged theoretical maximum water-recovery factor, distillate mass flux, temperature-polarization coefficient, and DCMD module thermal-efficiency metrics, represented by GOR and SEC_{th}. Three heat-recovery configurations were evaluated for two feed salinity levels and different feed and distillate flow rates.

The results show that heat recovery improves the energetic performance under all the evaluated conditions. For $w_{NaCl,s} = 3.5\%$, operation close to the critical flow-rate ratio, $\alpha = \alpha^* = 0.88$, reaches GOR = 3.3 and $\text{SEC}_{th} = 0.21 \text{ kWh L}^{-1}$ when both heat exchangers operate with an effectiveness of 0.8.

Increasing the feed salinity to $w_{NaCl,s} = 12.0\%$ reduces both productivity and thermal efficiency because the vapor-pressure driving force decreases. Nevertheless, the use of both heat exchangers still provides the highest GOR and the lowest SEC_{th} among the evaluated heat-recovery configurations.

Figures 8 and 9 illustrate the dynamic behavior of GOR and SEC_{th} under batch operation. The two maximum allowable salt concentrations considered in the mixing tank are $C_{s,\max} = 120 \text{ g}_\text{s} \text{ L}_\text{mix}^{-1}$ and $C_{s,\max} = 260 \text{ g}_\text{s} \text{ L}_\text{mix}^{-1}$.The average

Table 6: Influence of the operational conditions on the averaged DCMD module efficiency metrics under steady-state operation over 8 h.

Parameter	$w_{NaCl,s} = 3.5\%$			$w_{NaCl,s} = 12.0\%$	
	Condition 1 $\dot{V}_l = 30$ $\dot{V}_s = 80$	Condition 2 $\dot{V}_l \approx 44$ $\dot{V}_s = 50$	Condition 3 $\dot{V}_l = 50$ $\dot{V}_s = 50$	Condition 4 $\dot{V}_l \approx 42$ $\dot{V}_s = 50$	Condition 5 $\dot{V}_l = 80$ $\dot{V}_s = 80$
Flow-rate relation	$\dot{V}_l < \dot{V}_s$	$\dot{V}_l < \dot{V}_s$	$\dot{V}_l = \dot{V}_s$	$\dot{V}_l < \dot{V}_s$	$\dot{V}_l = \dot{V}_s$
α	0.38	0.88	1.00	0.74	0.88
α^*	0.88	0.88	0.88	0.74	0.74
Operating regime	DLR	FLR	FLR	FLR	FLR
T_{thr} [°C]	80.0	—	—	—	—
Z_{thr} [m]	0.63	—	—	—	—
RR [%]	2.5	5.0	5.2	2.2	2.9
$\dot{V}_{well}$ [L/min]	2.6	3.3	3.4	1.7	3.6
TPC [-]	0.73	0.56	0.59	0.72	0.68
η_{th} [%]	73	67	66	37	50
GOR/SEC$_{th}$ *values*					
No heat recovery: $\varepsilon_{HX1} = \varepsilon_{HX2} = 0$	0.28 / 2.5	0.56 / 1.2	0.58 / 1.2	0.30 / 2.3	0.40 / 1.7
HX-1 only: $\varepsilon_{HX1} = 0.8$, $\varepsilon_{HX2} = 0$	0.41 / 1.7	1.9 / 0.36	1.8 / 0.38	0.88 / 0.79	0.96 / 0.72
HX-1 and HX-2: $\varepsilon_{HX1} = \varepsilon_{HX2} = 0.8$	1.4 / 0.48	3.3 / 0.21	2.5 / 0.27	1.6 / 0.43	1.5 / 0.46

The values in the three final rows are presented as GOR/SEC$_{th}$, with SEC$_{th}$ expressed in kWh/L.
(a) Feed-side threshold temperature, $T_{thres,s}$.
(b) Distillate-side threshold temperature, $T_{thres,l}$.

values indicated in these figures were calculated over an operating period of 8 h.

Figure 10 illustrates the temporal evolution of the main DCMD system parameters under the different desalination-system operating modes: batch, semi-batch, volume control, continuous operation, and steady-state operation. The evaluated parameters are the salt concentration in the mixing tank, $C_{s,mix}$; the volumetric flow rate of produced distillate, $\dot{V}_{jwl}$; the mixture volume in the mixing tank, V_{mix}; the mixture temperature, T_{mix}; the rejection fraction, λ; and the volumetric flow rate supplied by the brackish-water well, $\dot{V}_{well}$.

According to Figure 10, the cycle duration is approximately 1789 s ($\approx$ 30 min) for batch operation, 4800 s ($\approx$ 1 h 20 min) for volume-control operation, and 6090 s ($\approx$ 1 h 41 min) for semi-batch operation.To identify the most appropriate operating mode for the DCMD desalination system, Figures 11 and 12 compare the average specific thermal energy consumption, SEC$_{th}$, and the

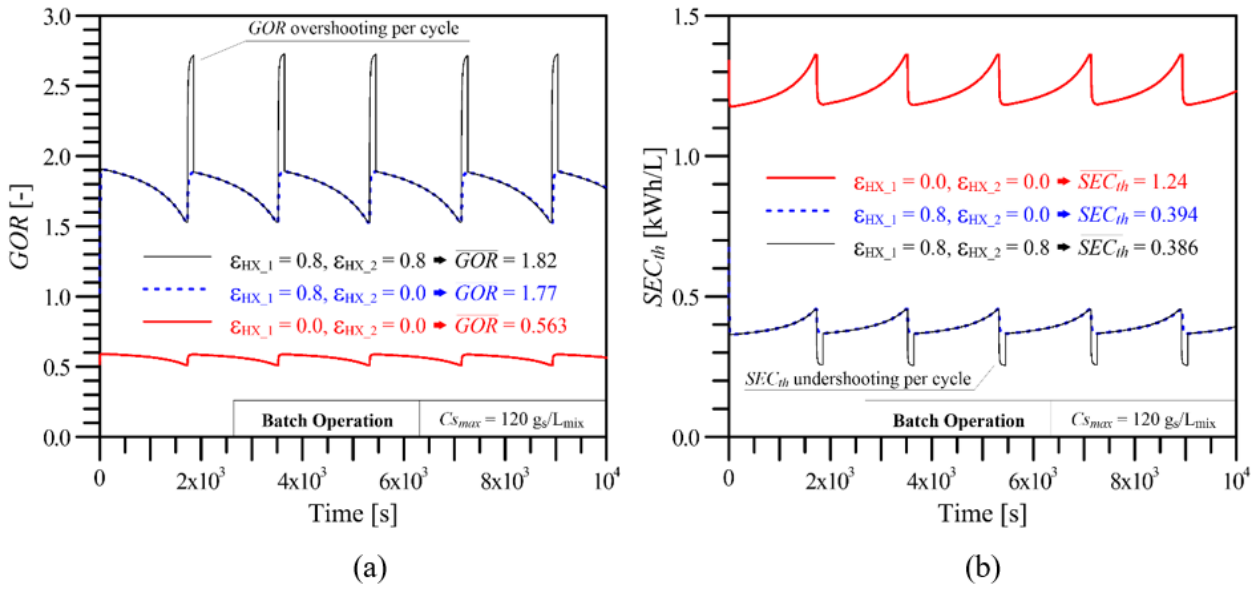


Figure 8: Evolution of the instantaneous (a) GOR and (b) SEC_{th} for $C_{s,\max} = 120\ \mathrm{g_s\,L_{mix}^{-1}}$ under batch operation.

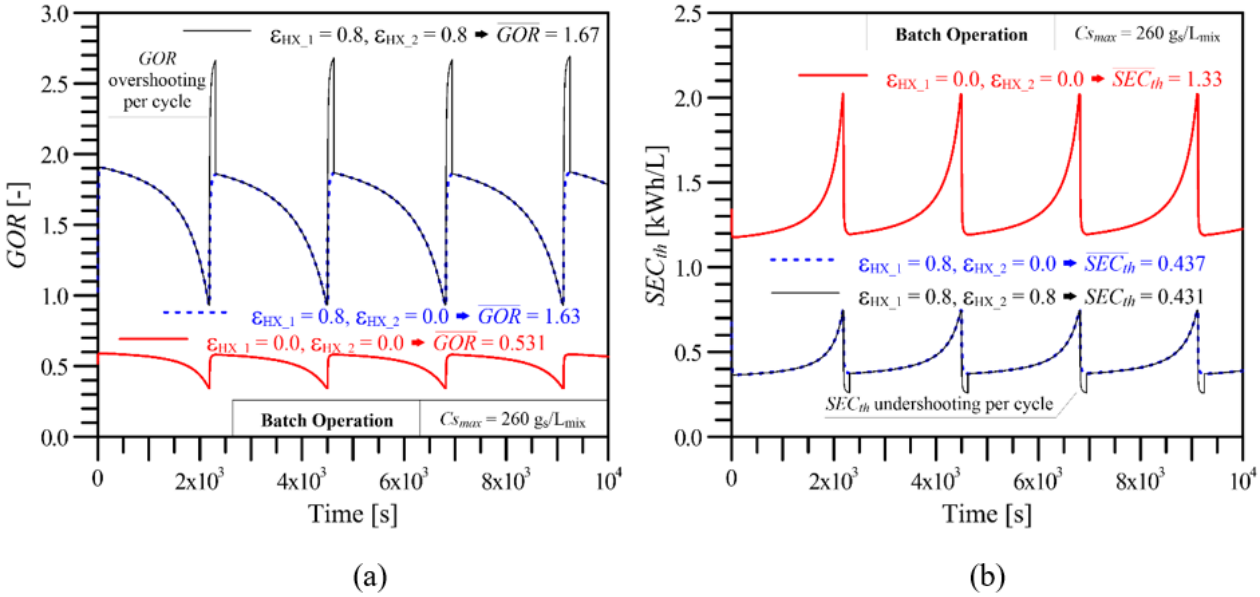


Figure 9: Evolution of the instantaneous (a) GOR and (b) SEC_{th} for $C_{s,\max} = 260\ \mathrm{g_s\,L_{mix}^{-1}}$ under batch operation.

water recovery ratio, RR, respectively.

Overall, these results indicate that a balance must be sought between specific energy consumption and water recovery, since a trade-off exists between these metrics for the same maximum operating salinity. Regarding energy consump-

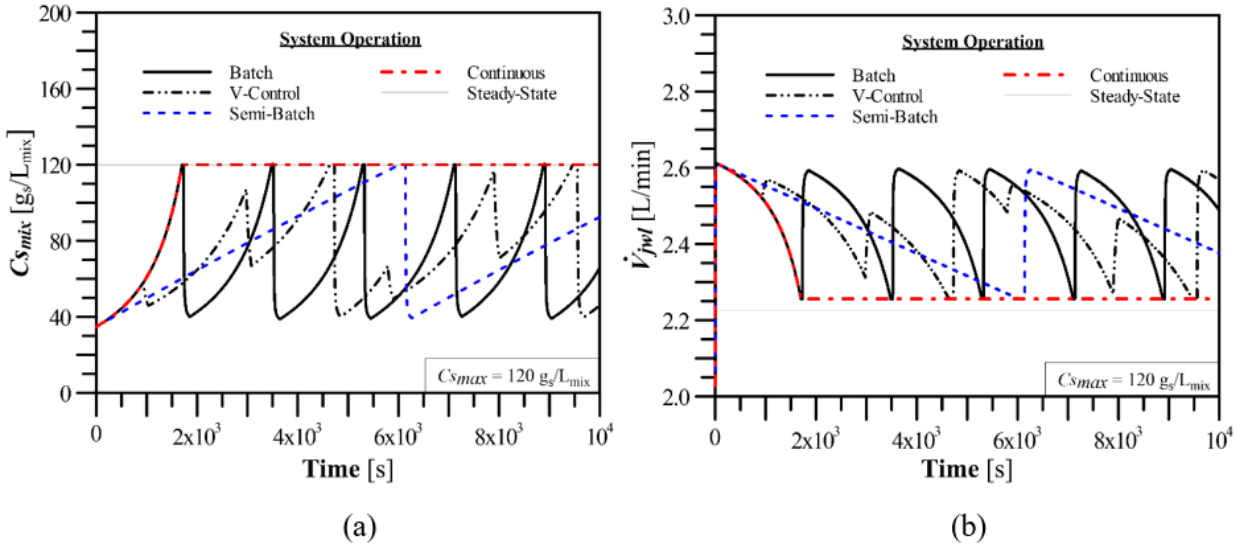


Figure 10: Dynamic evolution of the instantaneous desalination-system parameters under different operating modes for $C_{s,\max} = 120\ \mathrm{g_s\,L_{mix}^{-1}}$: (a) $C_{s,\mathrm{mix}}$ and (b) $\dot{V}_{jwl}$.

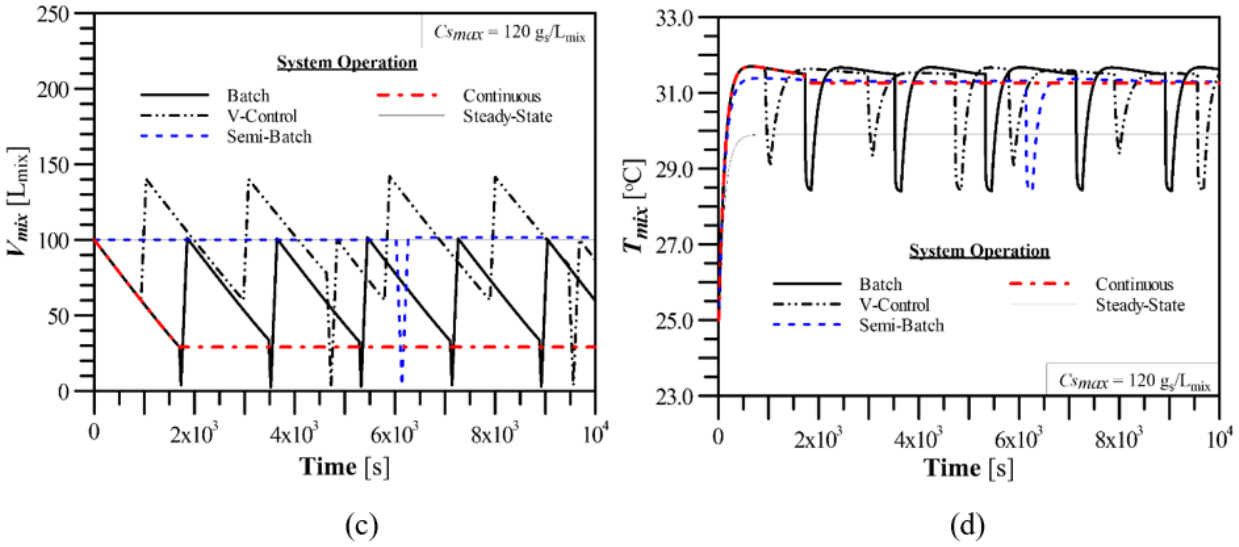


Figure 10: Dynamic evolution of the instantaneous desalination-system parameters under different operating modes for $C_{s,\max} = 120\ \mathrm{g_s\,L_{mix}^{-1}}$, continued: (c) V_{mix} and (d) T_{mix}.

tion, the batch mode presents the best performance, followed by the semi-batch, steady-state, and continuous modes, in agreement with the literature [Swaminathan and Lienhard, 2018]. Regarding water recovery, Figure 12 indicates that semi-batch operation provides the highest recovery ratio, followed by the

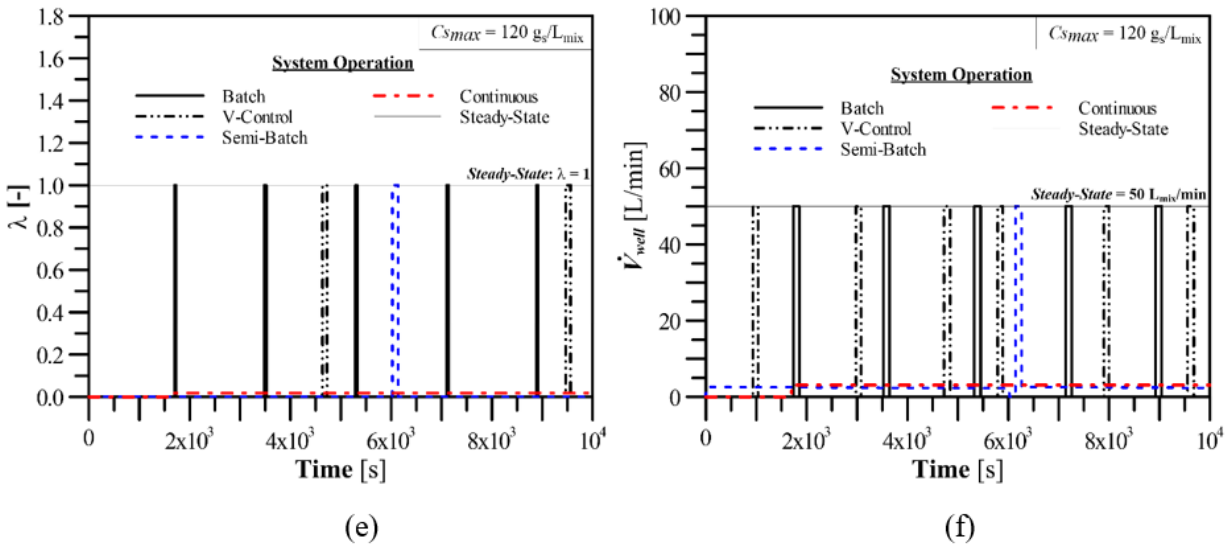


Figure 10: Dynamic evolution of the instantaneous desalination-system parameters under different operating modes for $C_{s,\max} = 120\ \mathrm{g_s\,L_{mix}^{-1}}$, continued: (e) λ and (f) $\dot{V}_{\mathrm{well}}$.

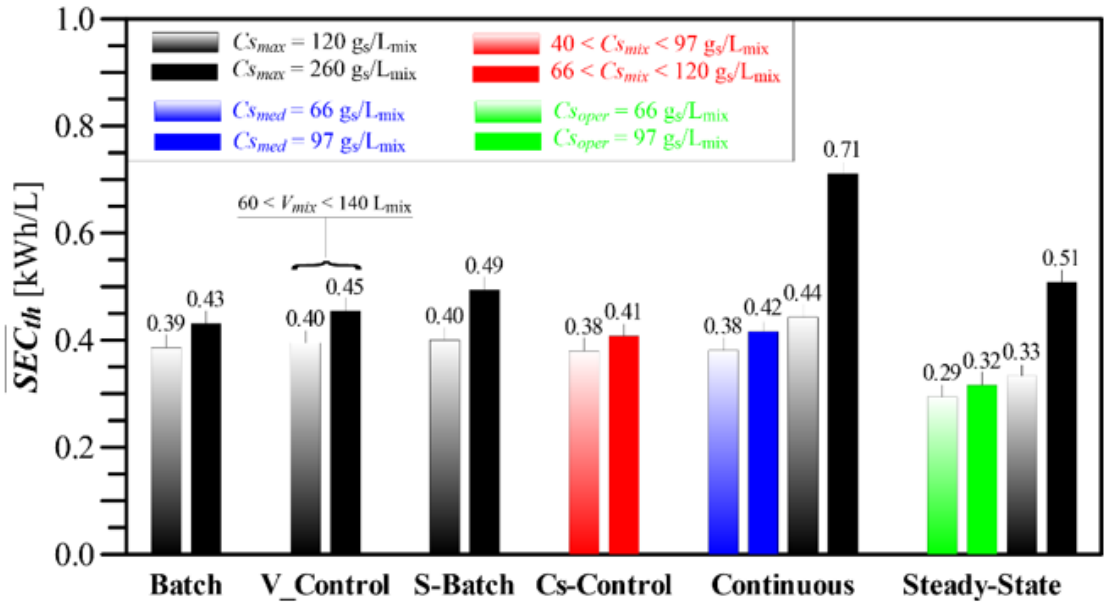


Figure 11: Effect of the system operating mode on the average specific thermal energy consumption, SEC_{th}, for different maximum mixture salinities.

volume-control and batch modes.

Figure 13 compares the volume of distillate produced under each operating mode over an 8 h operating period.

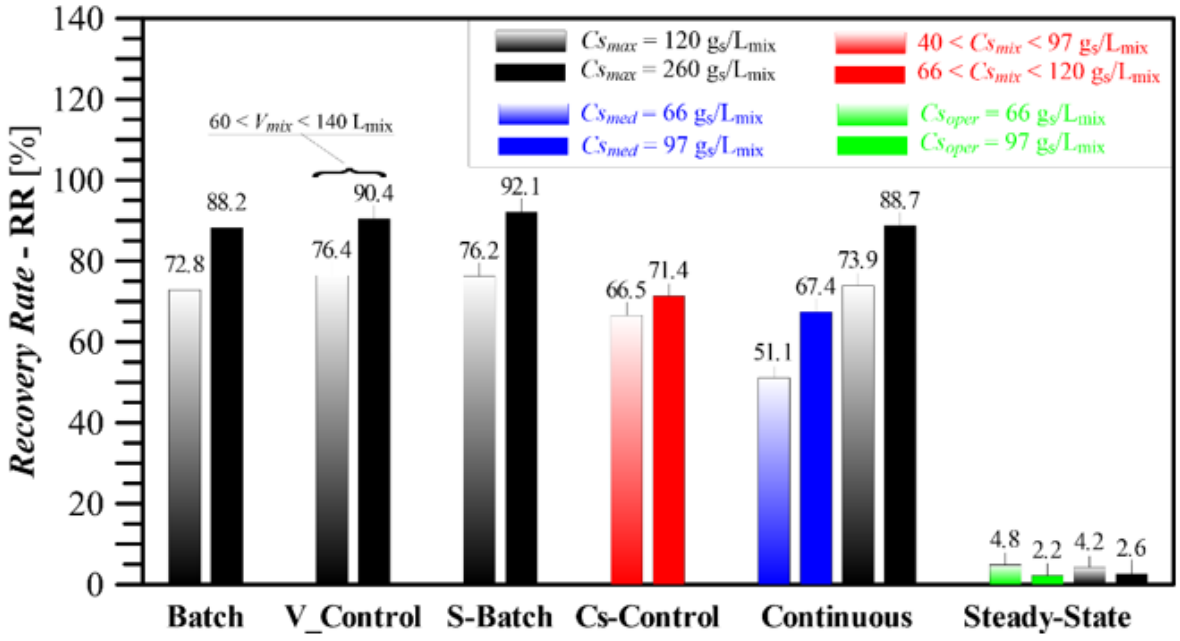


Figure 12: Effect of the system operating mode on the water recovery ratio, RR, for different maximum mixture salinities.

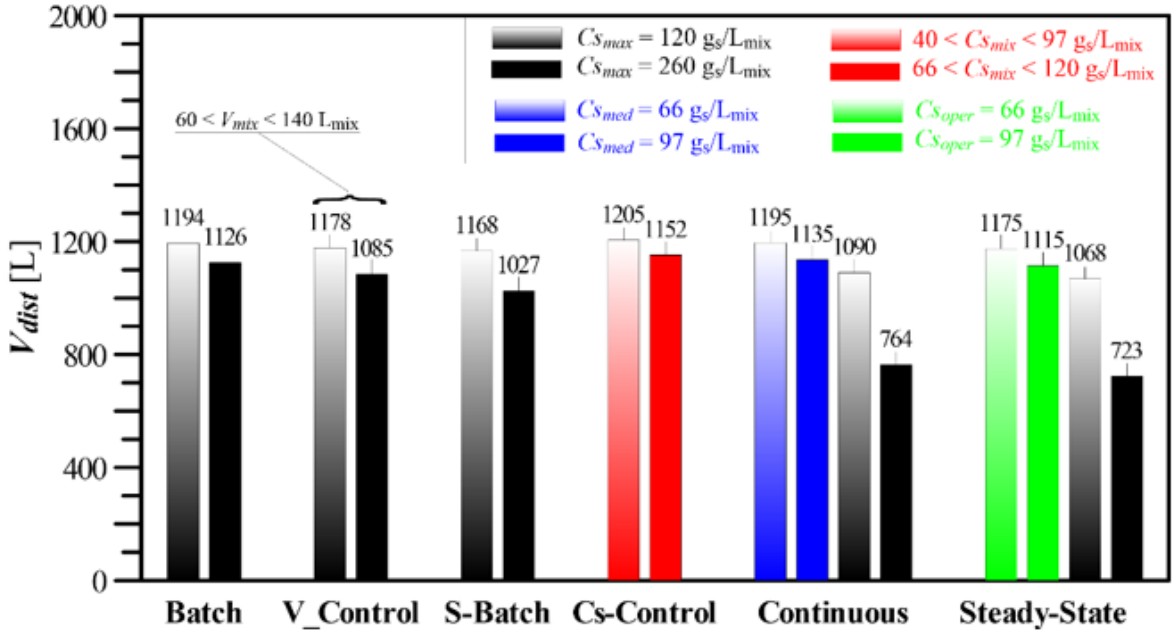


Figure 13: Effect of the system operating mode on the volume of distillate produced over 8 h for different maximum mixture salinities.

For the lower salinity level, the desalination system can produce approximately 1,200 L of distillate over 8 h under any of the evaluated operating modes. Increasing salinity reduces distillate production, particularly under the continuous and steady-state modes, for which approximately 750 L of distillate is produced.

Therefore, for applications in regions subjected to seasonal water stress, such as the semi-arid region of northeastern Brazil, semi-batch operation may be the preferred alternative for brackish-water desalination because it provides the best compromise when minimizing well-water consumption is the main objective.

6 Conclusions

The present work developed a mathematical modeling and computer simulation of a DCMD desalination system with double heat recovery from the distillate stream and the feed stream, together with total or partial recirculation of the retentate. The main goal was to employ the developed code to identify the best scenarios for operation of the system in terms of maximizing its thermal efficiency while simultaneously minimizing the use of brackish water from wells, seeking optimized production of the amount of distilled water.

The results show that heat recovery from the distillate stream already produces a significant increase in the energy efficiency of the desalination system, constituting a mandatory condition for the operation of DCMD-based systems. Beyond this, the use of an additional heat exchanger to recover heat from the retentate stream further improves the energy efficiency of the process, an effect that is especially pronounced when the system operates in the FLR condition with a mass flow rate near the critical mass flow rate. Under these circumstances, peaks in GOR values and lows in SEC values can be observed when the well and an external heat exchanger coupled to its stream operate simultaneously. Double heat recovery and partial recirculation of the retentate therefore play an important role in the competitiveness of the membrane distillation pro-

cess compared to other forms of desalination, particularly in long operation processes.

The system behavior also proved sensitive to two operational factors. The amount of heat supplied to the system is strongly impacted by the rejection fraction of the retentate stream, and the operating mode itself has a strong influence on both the energy consumption and the water recovery of the DCMD desalination system. In this regard, the lowest specific thermal energy cost is associated with the batch operation mode of the desalination plant. However, in regions subject to hydrologic seasonal stress, the results indicate that the semi-batch mode could be the preferred choice for brackish water saving. The selection of an operating mode must consequently be guided by site-specific information, such as where the system is to be implemented and what the local availability of energy and brackish water salinity are.

Extension to other desalination MD methods, such as AGMD and V-AGMD, is currently under development, in order to simulate the entire experimental TRL-6 desalination unit installed in the LabMEMS/UFRJ.

References

S. Al-Obaidani, E. Curcio, F. Macedonio, G. Di Profio, H. Al-Hinai, and E. Drioli. Potential of membrane distillation in seawater desalination: thermal efficiency, sensitivity study and cost estimation. *Journal of Membrane Science*, 323(1):85–98, 2008.

A. Alkhudhiri, N. Darwish, and N. Hilal. Membrane distillation: A comprehensive review. *Desalination*, 287:2–18, 2012.

J. Beeckman. Mathematical description of heterogeneous materials. *Chemical Engineering Science*, 45:2603–2610, 1990.

C. Charcosset. A review of membrane processes and renewable energies for desalination. *Desalination*, 245:214–231, 2009.

K. S. S. Christie, T. Horseman, and S. Lin. Energy efficiency of membrane distillation: simplified analysis, heat recovery and the use of waste-heat. *Environment International*, 138:105588, 2020.

E. Curcio and E. Drioli. Membrane distillation and related operations – a review. *Separation and Purification Reviews*, 34:35–86, 2005.

E. Drioli, A. Ali, and F. Macedonio. Membrane distillation: Recent developments and perspectives. *Desalination*, 356:56–84, 2015.

T. Elias-Kohav, S. Moshe, and D. Avnir. Steady-state diffusion and reactions in catalytic fractal porous media. *Chemical Engineering Science*, 46(11):2787–2798, 1991.

M. Elimelech and W. A. Phillip. The future of seawater desalination: Energy, technology and the environment. *Science*, 333:712–717, 2011.

H. Fan and Y. Peng. Application of symmetric PVDF membranes in desalination and comparison of the VMD and DCMD processes. *Chemical Engineering Science*, 79:94–102, 2012.

B. Ghanbarian, A. G. Hunt, R. P. Ewing, and M. Sahimi. Tortuosity in porous media: a critical review. *Soil Science Society of America Journal*, 77:1461–1477, 2013.

J. Gilron, L. M. Song, and K. K. Sirkar. Design of cascade of crossflow direct contact membrane distillation. *Industrial & Engineering Chemistry Research*, 46(8):2324–2334, 2007.

R. Gonzalez-Bravo, N. A. Elsayed, J. M. Ponce-Ortega, F. Napoles-Rivera, and M. M. El-Halwagi. Optimal design of thermal membrane distillation systems with heat integration with process plants. *Applied Thermal Engineering*, 75: 154–166, 2015.

C. Gostoli and G. C. Sarti. Low temperature distillation through hydrophobic membrane. *Separation Science and Technology*, 22(2–3):855–872, 1998.

G. Guan, X. Yang, R. Wang, R. Field, and A. G. Fane. Evaluation of hollow fiber-based direct contact and vacuum membrane distillation systems using Aspen process simulation. *Journal of Membrane Science*, 464:127–139, 2014.

G. Guan, X. Yang, R. Wang, and A. G. Fane. Evaluation of heat utilization in membrane desalination system integrated with heat recovery. *Desalination*, 366:80–93, 2015.

A. Hausmann, P. Sanciolo, T. Vasiljevic, M. Weeks, and M. Duke. Integration of membrane distillation into heat paths of industrial processes. *Chemical Engineering Journal*, 211–212:378–387, 2012.

IMSL Fortran Math Library. *IMSL Fortran Math Library*. Perforce/Rogue Wave Software, Minneapolis, 2022.

M. S. Khayet and T. Matsuura. *Membrane Distillation: Principles and Applications*. Elsevier, Amsterdam, 2011.

M. S. Khayet, P. Godino, and J. I. Mengual. Theory and experiments on sweeping gas membrane distillation. *Journal of Membrane Science*, 165:261–272, 2000.

M. S. Khayet, T. Matsuura, J. I. Mengual, and M. Qtaishat. Design of novel direct contact membrane distillation membranes. *Desalination*, 192(1–3):105–111, 2006.

A. S. Kim. Cylindrical cell model for direct contact membrane distillation (DCMD) of densely packed hollow fibers. *Journal of Membrane Science*, 455: 168–186, 2014.

W.-J. Kim, O. Campanella, and D. R. Heldman. Predicting the performance of direct contact membrane distillation (DCMD): Mathematical determination

of appropriate tortuosity based on porosity. *Journal of Food Engineering*, 294:110400, 2021.

Y. D. Kim, K. Thu, and S. H. Choi. Solar-assisted multi-stage vacuum membrane distillation system with heat recovery unit. *Desalination*, 367:161–171, 2015.

H. Y. Lee, F. He, L. M. Song, and K. K. Sirkar. Desalination with a cascade of crossflow hollow fiber membrane distillation devices integrated with a heat exchanger. *AIChE Journal*, 54(7):1780–1795, 2011.

Z. Li, Y. Peng, Y. Dong, H. Han, P. Chen, L. Qiu, and Q. Jiang. Effects of thermal efficiency in DCMD and the preparation of membranes with low thermal conductivity. *Applied Surface Science*, 317:338–349, 2014.

S. Lin, N. Y. Yip, and M. Elimelech. Direct contact membrane distillation with heat recovery: thermodynamic insights from module scale modeling. *Journal of Membrane Science*, 453:498–515, 2014.

K. M. Lisboa, D. B. Moraes, C. P. Naveira-Cotta, and R. M. Cotta. Analysis of the membrane effects on the energy efficiency of water desalination in a direct contact membrane distillation (DCMD) system with heat recovery. *Applied Thermal Engineering*, 182:116063, 2021.

Y. Lu and J. Chen. Integration design of heat exchanger networks into membrane distillation systems to save energy. *Industrial & Engineering Chemistry Research*, 51:6798–6810, 2012.

K. S. Maheswari, K. Kalidasa Murugavel, and G. Esakkimuthu. Thermal desalination using diesel engine exhaust waste heat – an experimental analysis. *Desalination*, 358:94–100, 2015.

P. A. B. Sampaio. Computational model and simulation of DCMD desalination systems with heat recovery. *Desalination*, 533:115769, 2022.

M. H. Sharqawy, J. H. Lienhard, and S. M. Zubair. The thermophysical properties of seawater: A review of existing correlations and data. *Desalination and Water Treatment*, 16:354–380, 2010.

B. S. Sparrow. Empirical equations for the thermodynamic properties of aqueous sodium chloride. *Desalination*, 159:161–170, 2003.

J. Swaminathan and J. H. Lienhard, V. Design and operation of membrane distillation with feed recirculation for high recovery brine concentration. *Desalination*, 445:51–62, 2018.

J. Swaminathan, H. W. Chung, D. M. Warsinger, and J. H. Lienhard V. Energy efficiency of membrane distillation up to high salinity: Evaluating critical system size and optimal membrane thickness. *Applied Energy*, 211:715–734, 2018.

H. P. Veiga. Encontros temáticos – saneamento ambiental: Desafios e soluções. Technical report, ABES-RS, Ministério do Meio Ambiente, Porto Alegre/RS, Brazil, 2017.

K. Y. Wang, T. S. Chung, and M. Gryta. Hydrophobic PVDF hollow fiber membranes with narrow pore size distribution and ultra-thin skin for the fresh water production through membrane distillation. *Chemical Engineering Science*, 63(9):2587–2594, 2008.

K. Y. Wang, S. W. Foo, and T. S. Chung. Mixed matrix PVDF hollow fiber membranes with nanoscale pores for desalination through direct contact membrane distillation. *Industrial & Engineering Chemistry Research*, 48:4474–4483, 2009.

X. Yang, R. Wang, L. Shi, A. G. Fane, and M. Debowski. Performance improvement of PVDF hollow fiber-based membrane distillation process. *Journal of Membrane Science*, 369:437–447, 2011.

K. Yao, Y. Qin, and Y. Yuan. A continuous-effect membrane distillation process based on hollow fiber AGMD module with internal latent-heat recovery. *AIChE Journal*, 59:1278–1297, 2013.

J. Zhang, J. Li, M. Duke, Z. Xie, and S. Gray. Performance of asymmetrical hollow fiber membranes in membrane distillation under various configurations and vacuum enhancement. *Journal of Membrane Science*, 362:517–528, 2010.

Y. Zhang, Y. Peng, S. Ji, Z. Li, and P. Chen. Review of thermal efficiency and heat recycling in membrane distillation processes. *Desalination*, 367:223–239, 2015.